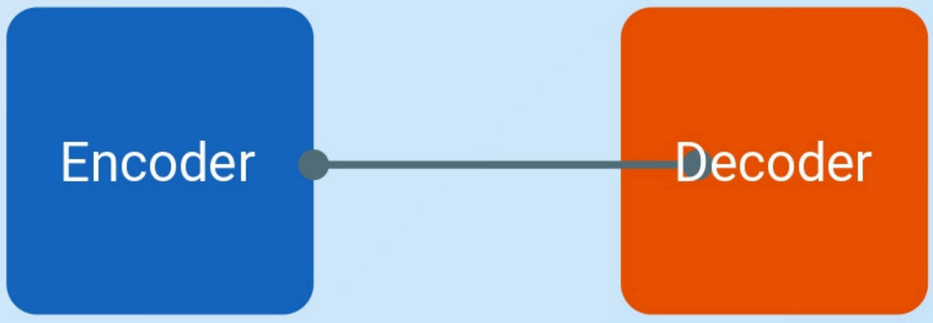


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Understanding Transformer Architecture

Exploring Encoder-Only, Decoder-Only, and Encoder-Decoder Models



Encoder-Only

Understanding & Analysis

Decoder-Only

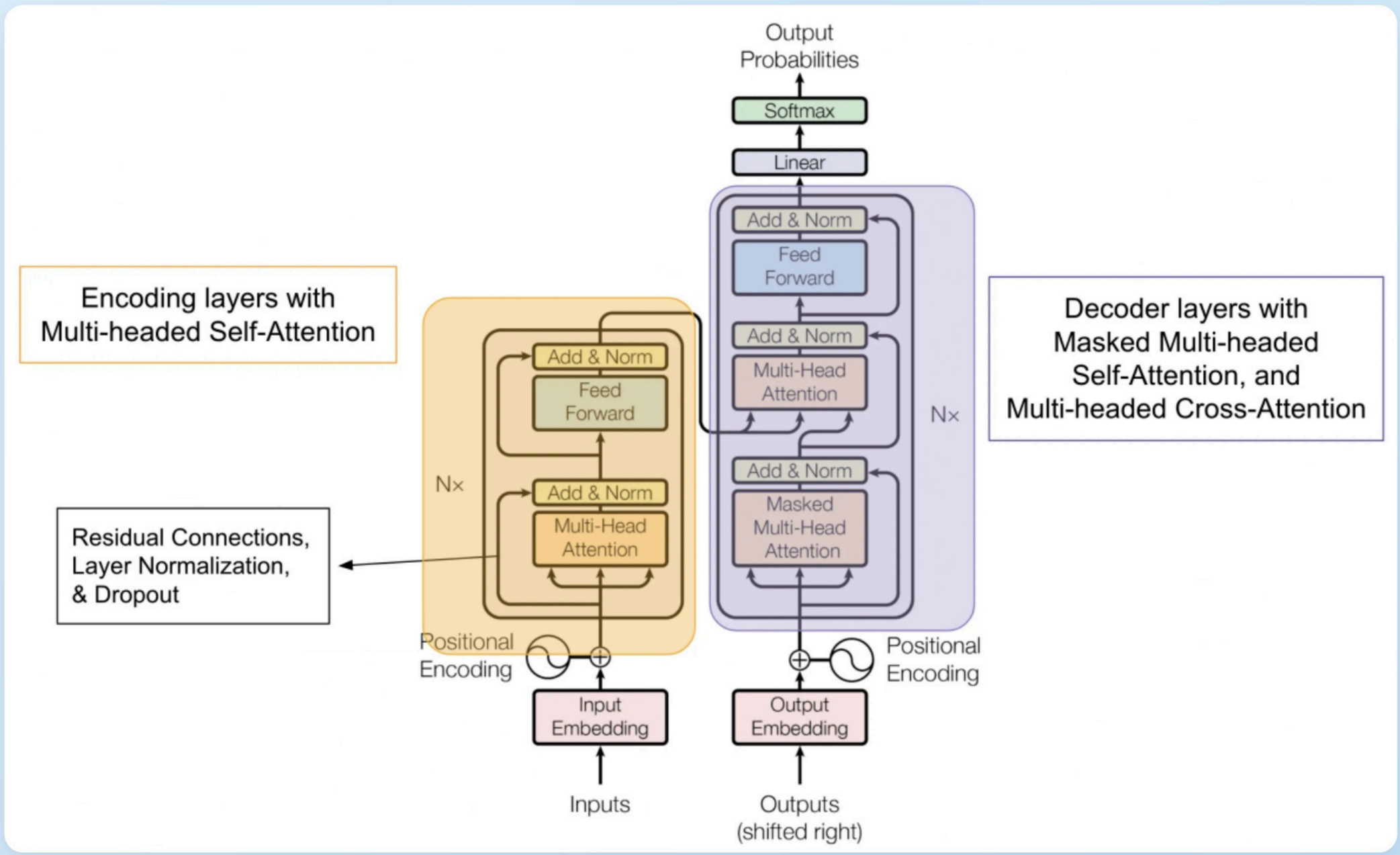
Generation & Creation

Encoder-Decoder

Transformation & Mapping

Transformer Architecture Overview

The transformer architecture revolutionized NLP with its self-attention mechanism, enabling parallel processing and capturing long-range dependencies.



Self-Attention

Mechanism that allows each position to attend to all positions in the previous layer. Computes **weighted sums** of values based on compatibility between query and key vectors.

Encoder-Decoder

Stack of identical layers. **Encoder** processes input bidirectionally, **Decoder** generates output using encoder information and previously generated tokens.

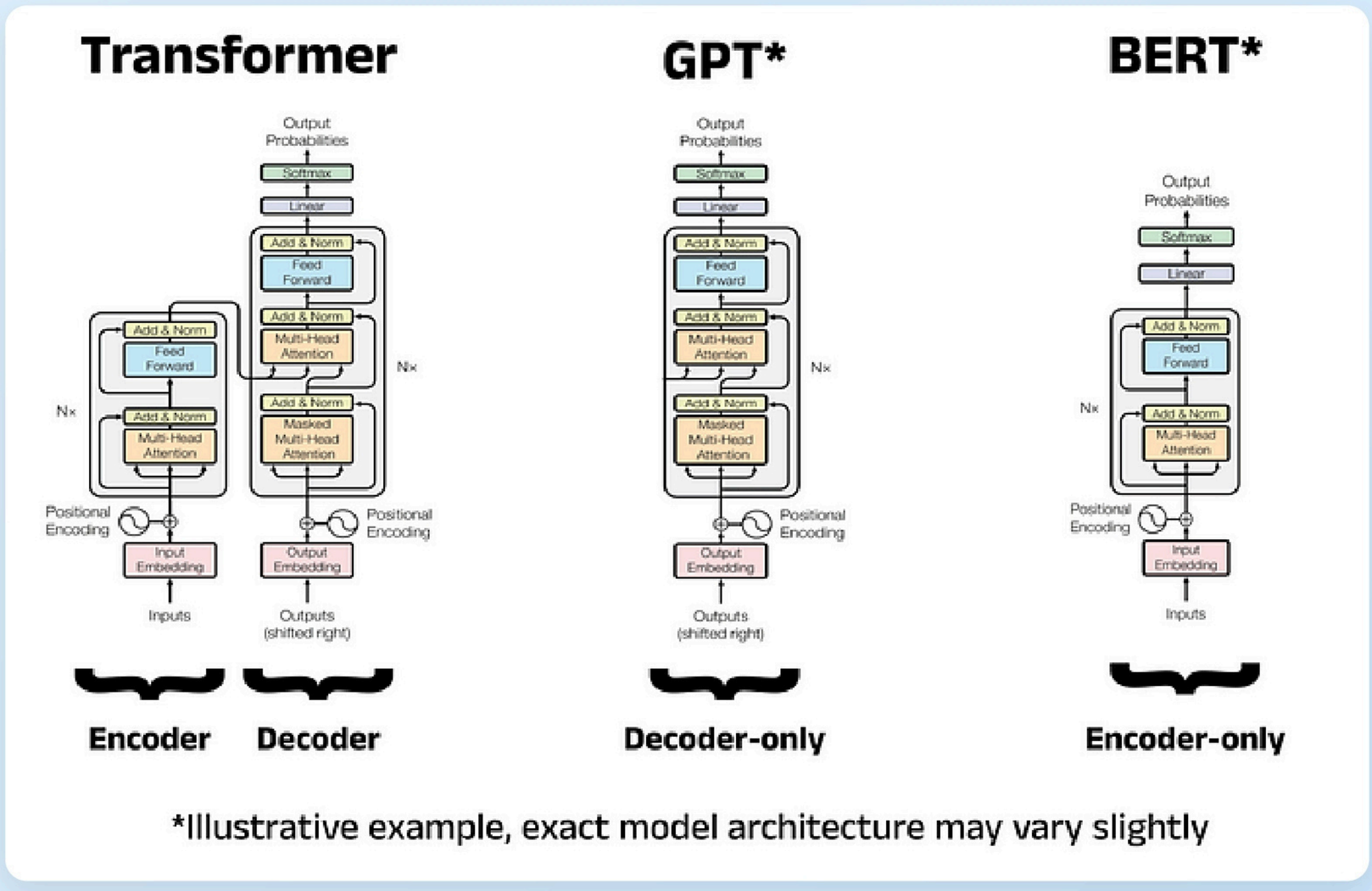
Positional Encoding

Adds information about the position of tokens in the sequence using **sine and cosine functions** of different frequencies. Compensates for the lack of recurrence/convolution.

Multi-Head Attention

Allows the model to **jointly attend** to information from different representation subspaces at different positions. Enables learning of various types of relationships.

Encoder-Only Transformer Models



*Illustrative example, exact model architecture may vary slightly

Bidirectional Attention

Processes input tokens in **both directions** simultaneously, allowing each token to attend to all other tokens in the sequence. Captures full context from preceding and following words.

Training Methods

Uses **Masked Language Modeling** (MLM) where random tokens are masked and predicted, and **Next Sentence Prediction** to understand relationships between sentences.

Architecture Details

Stack of identical encoder layers with multi-head self-attention and position-wise feed-forward networks. No decoder component. Outputs rich contextual embeddings.

How It Works

Input tokens are embedded, positional encodings are added, then processed through multiple encoder layers. The final hidden states capture **contextual representations** of each token.

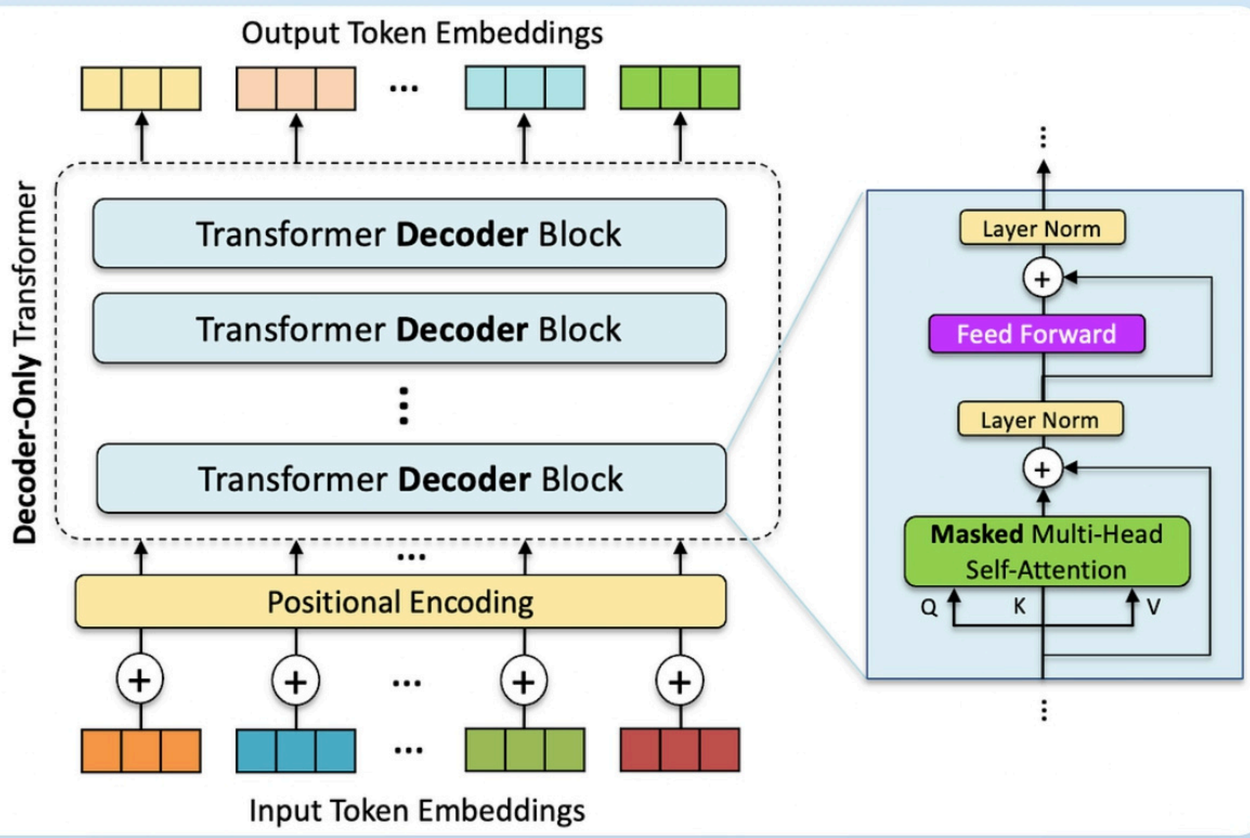
Key Use Cases & Applications

- Sentiment Analysis
- Named Entity Recognition
- Text Classification
- Question Answering

BERT · RoBERTa · DistilBERT · ALBERT

Decoder-Only Transformer Models

GPT



Causal Masking

Uses **masked self-attention** that restricts each position to only attend to previous positions in the sequence. Prevents "peeking" at future tokens during generation.

Training Methods

Trained using **autoregressive language modeling** - predicting the next token in a sequence given all previous tokens. Maximizes likelihood of observed text data.

Architecture Details

Stack of identical decoder layers with masked multi-head self-attention and position-wise feed-forward networks. No encoder component. Generates text **sequentially**.

How It Works

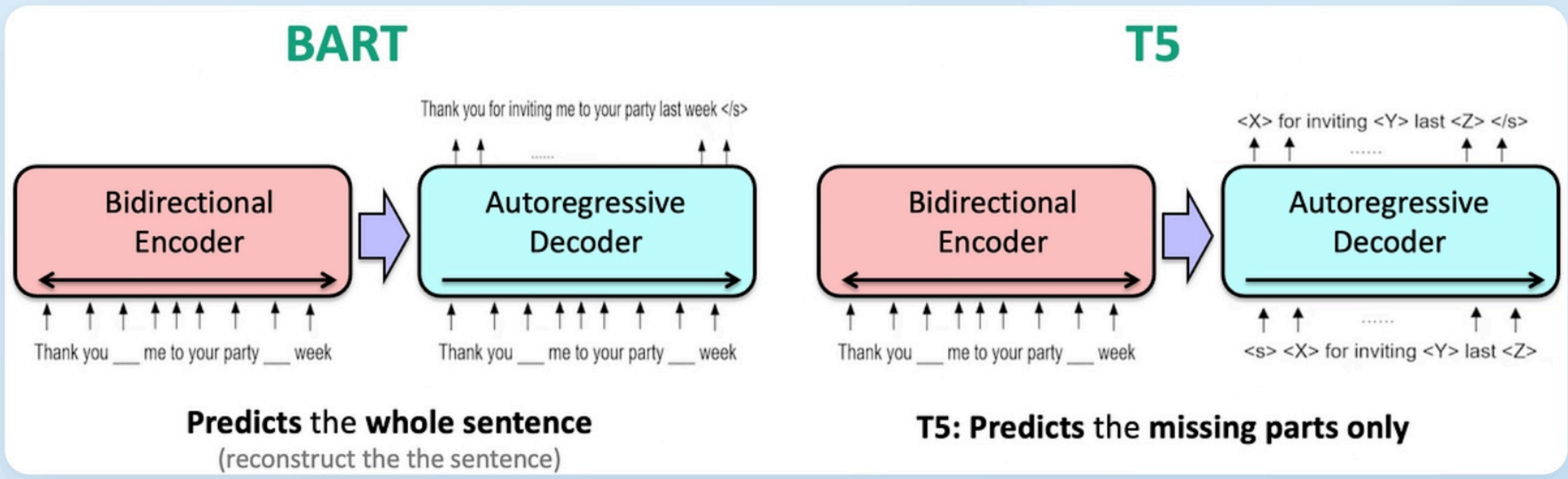
Input tokens are embedded, positional encodings are added, then processed through multiple decoder layers. Each token prediction is conditioned on **all previous tokens** in the sequence.

Key Use Cases & Applications

- Text Generation
- Chatbots
- Code Writing
- Story Completion

GPT · GPT-2 · GPT-3 · GPT-4 · LLaMA

Encoder-Decoder Transformer Models



Dual Component Architecture

Combines both **encoder and decoder** stacks. Encoder processes input bidirectionally while decoder generates output sequentially using causal masking.

Cross-Attention Mechanism

Decoder uses **cross-attention** to attend to encoder outputs, allowing it to focus on relevant parts of the input sequence when generating each output token.

Training Methods

Trained on **sequence-to-sequence** tasks using teacher forcing. Models like T5 use text-to-text framework, while BART uses denoising autoencoding objectives.

How It Works

Input sequence is encoded into contextual representations, then decoder generates output sequence **token by token**, attending to both previously generated tokens and encoder outputs.

Key Use Cases & Applications

- Language Translation
- Text Summarization
- Question Answering
- Text-to-SQL

T5 · BART · Original Transformer · Pegasus

Transformer Architecture Comparison

Model Type	Architecture	Training Approach	Strengths & Weaknesses
Encoder-Only	Stack of encoder layers with bidirectional self-attention. Processes entire input simultaneously.	Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). Predicts masked tokens.	**Strengths:** Excellent understanding tasks, rich contextual embeddings. **Weaknesses:** Not designed for text generation.
Decoder-Only	Stack of decoder layers with causal masking. Processes tokens left-to-right sequentially.	Autoregressive language modeling. Predicts next token given previous tokens in sequence.	**Strengths:** Superior text generation, versatile applications. **Weaknesses:** Can only see past context, not future.
Encoder-Decoder	Encoder processes input bidirectionally, decoder generates output with cross-attention to encoder.	Sequence-to-sequence training with teacher forcing. Maps input sequences to output sequences.	**Strengths:** Ideal for transformation tasks, handles different input/output formats. **Weaknesses:** More complex, higher computational cost.

Choosing the Right Architecture

What's your primary task?	Need bidirectional context?	Input and output formats?
Understanding Text	Yes	Same
Generating Text	Sometimes	Same
Transforming Text	No	Different

Encoder-Only Success

- Google Search BERT integration
- Twitter sentiment analysis
- Medical text classification

Decoder-Only Success

- ChatGPT conversational AI
- GitHub Copilot code generation
- AI content creation tools

Encoder-Decoder Success

- Google Translate
- News article summarization
- Multilingual question answering